

Assignment: Write a one-page resume. (The one-page limit is required.)

Due: Draft 1 is due NLT 2359 Monday 8 September 2026

Point Value: 200

Your task is to write a one-page resume that demonstrates (1) professional engagement, (2) recognition of chemical engineering skills, (3) intellectual growth in chemical engineering, and (4) written communication skills. Your resume will be assessed with the rubric at this [link](#). Read the rubric carefully since the levels will be rigorously applied to determine your grade.

A resume is a brief one-page brag sheet providing a concise overview of your skills and qualifications and normally includes relevant experience, skills, and education. Along with your contact information, these topics also form the main sections of the resume. A description of what to include in the sections is in enclosure 1.

Well-written resumes demonstrate introspection and self-awareness because they are a personal inventory of professional skills and experience. *The purpose of this assignment is for you to reflect on your skills and experiences and describe them in a resume.*

A resume highlights your skills as a chemical engineer in a way that relates to the reader. While the reader is normally a potential employer, finding a job is not the purpose of this assignment. However, you are strongly encouraged to role-play as if you are. This will help you to decide what is relevant by giving you a target audience. Two job postings are included in enclosures 2 and 3, and enclosure 4 has some pre-approved web sites with additional postings.

LyX (pronounced like “licks”) is required for this assignment. An instructor-provided LyX template can be found in CANVAS. The template is also required to save time formatting your resume. LyX is a front end for LaTeX (pronounced “LAY-tek”), which is a professional programming language for scientific and mathematical type-setting. LyX encourages an approach to writing based on the content of your document and not simply appearance, allowing you to write structured, high-quality documents without worrying about low-level formatting. The software is already installed on the lab computers in 331, but you may also want to install it on your laptops. Instructions for installing LyX are found in enclosure 5.

Feel free to use any appropriate internet resources to help you craft your resume. Generative AI is a very good resource. For example, using the prompt “Can you generate a 1-page resume for a military cadet seeking an entry-level job in chemical engineering at DuPont?” gives a good result. Of course, your instructor is aware of AI-generated resumes so be very careful when using this tool to craft your resume to your personal qualifications and skills.

Some of you will be working on CVs for other courses. This exercise is not to write a CV and CVs are not resumes. A CV or Curriculum Vitae is a detailed chronological listing of prior employment experience and accomplishments, and does not have a page limit. CVs are normally used in academic settings to display credentials or to find academic jobs or scholarships.

Enclosure 1: What to Include in the Sections of the Resume

Summary Section

The summary section is normally optional but in this assignment it is required. Include valuable or relevant experience that does not fit in your experience and skills sections below. Briefly summarize your personality attributes that make you a strong candidate for the position, along with the title and advertisement reference number to which you are applying.

Pre-Professional and College Experience Section

It is best to write this section using projects completed in courses. Include the title of the project, the course number and title, and a description of relevant skills you learned in the project. For example, saying that you simulated a crude oil distillation column is so-so. It would be improved by adding details of how you built the column in CHEMCAD.

Add additional projects as space and experience allows.

Relevant coursework is a space-filler and is not in the grading rubric except indirectly through the communication category. Add this bullet to fill out white space in the resume and delete it if you need to add more project content.

Add an AIAD if you did one, including the organization, address, date, supervisor's name, and a description of what you did that emphasizes skills.

Cadet military experience is important. Add any jobs or tasks in USCC and describe any skills, tactical, administrative, leadership, and so on, that you displayed or learned in the job.

Other Skills and Qualifications Section

Describe your computing skills here. Knowledge of programming and simulation is best, for example, CHEMCAD and Mathematica, but don't just list software. Provide context by saying what you can do with the software. In other words, listing software would be a minimum grade, listing relevant software is better, and relevant software with skills and tasks is best.

Hobbies are optional. Add hobbies if space allows and only if they are relevant or good potential conversation starters in an interview.

Add SAcE or other relevant certifications.

Education Section

This is more or less the same for everyone. Add scholarships or academic awards if space allows.

Enclosure 2: Sample Job Posting from DuPont

Process Engineer (MAN01716)

Location: Hanko, Southern Finland, Finland Category Engineering Job

Job Type: Entry Level

Job Description

DuPont Nutrition & Biosciences is a USD 6.2 billion global business and one of the world's largest producers of specialty ingredients, developing and manufacturing solutions for the global food & beverage, health & bioscience and pharma markets. With more than 10,000 employees, 70+ manufacturing sites and 25+ R&D centers across the world, N&B is on a mission to provide essential innovations from highly engineered products and naturally sourced ingredients to shape industries and everyday life. More information can be found at <http://biosciences.dupont.com>.

DuPont Nutrition & Biosciences - Hanko plant is producing a variety of enzyme products by fermentation, recovery and formulation. We're looking for a PROCESS ENGINEER who believes that working together is the way to solve challenges and accomplish common goals.

Your responsibilities will include:

- Process performance and continuous optimization, troubleshooting and technical support for your process area.
- New product introductions to the plant together with global teams and R&D.
- Analysis of process data and compiling results in clear reports for global project teams.
- Process safety and sustainability in your process area are an integral part of your job.
- Investment project support and creating technical specifications for projects.
- Participating in process and equipment design, developing automation functional descriptions, and commissioning and start-up activities.
- You will participate in global knowledge teams that collaborate globally and drive technology development across production sites.

Qualifications

Ideally, you are a chemical or biochemical engineer with strong skills in process technology. You can be either an entry level candidate or a more senior expert, depending on the suitability & experience.

- M.Sc. or B.Sc. in Biochemical, Chemical or Process engineering.
- Previous working experience preferably from industry.
- Strong skills in process technology.
- Knowledge of energy technology (preferably) & statistical tools.
- Analytical, entrepreneurial, systematic and results driven mindset.
- Fluent oral and written skills in Finnish and English is required.
- Good IT skills.
- Willingness to travel occasionally.

Additionally, you prefer working together with people around you, you care and want to make processes safer and more efficient, and you like to work in a changing environment with speed and agility.

We offer you a permanent position in an exciting international company with opportunities to grow and develop, challenges where you need to utilize your technical knowledge & experience, and great opportunities to learn more.

Enclosure 3: Sample Job Posting from Ashland Chemical

Production Engineer (2026-028)

USA – Hopewell - VA

Ashland Specialty Ingredients, GP

Are you the kind of person that is always thinking, sketching, seeking, and adjusting? Who needs to understand how things work and then figure out how they can work better? Are you a passionate, tenacious, solver who loves to work with others who share your drive? Are you positive, constructive, and ingenious?

Are you always solving? Then we'd like to meet you and bet you'd like to meet us.

Ashland has an exciting opportunity for a Production Engineer to join our Specialty Ingredients business at our Hopewell, VA location. This is a very visible, significant role within the Company and the CMC function. This position will report to Operations Manager.

The responsibilities of the position include, but are not limited to, the following:

- Utilizes lean manufacturing techniques/tools for identification and implementation of continuous improvement projects.
- Analyzes production and quality data to determine optimum operating conditions.
- Utilize process techniques to meet customer requirements.
- Utilizes Lean manufacturing tools and techniques to improve product quality.
- Resolve production problems.
- Assist in creating and implementing new strategies to improve process efficiency.
- Provide on-time delivery of products.
- Maintain environmental compliance while continually pursuing waste minimization.
- Determines root cause of chronic operating problems.
- Provides technical support for capital projects including investigating new equipment.
- Investigates alternative raw materials for cost and quality improvements.
- Investigates opportunities to improve plant through-put and capacity.

In order to be qualified for this role, you *must possess* the following:

- 1-5 years' chemical plant manufacturing experience.
- Bachelor of Science Degree in Chemical Engineering.
- Must be authorized to work in the US.

The following skill sets are *preferred* by the business unit:

- Previous experience with Control Systems.
- Lean Six Sigma.
- Previous experience in Production Operations.
- Hazard Analysis - HAZOP, PHA, Good Manufacturing Practices, ISO Systems.
- Team Player with excellent written and oral communications.
- Proven data analysis and problem-solving skills.
- Proven planning and organization skills.
- Execute projects as part of a team, as a team leader or independently.
- Proven self-starter with urgency for delivering results.

Enclosure 4: Finding additional job postings

In role-playing, write your resume to address obtaining an internship or entry level position as a chemical or chemical process engineer. Pre-approved example job posting sites follow below:

- Go to the Eastman Chemical website at <https://www.eastman.com/Pages/Home.aspx> and click the menu button(), then click “Careers,” then click “Find Jobs.” In the “search by keyword” field, enter “chemical engineer” and then click the search jobs button (). Using the quotes around the search term will narrow the results considerably. Normally, several different chemical engineering jobs appear. Find an appropriate job posting to focus your resume. At the time this document was written, 15 chemical engineering jobs were posted here, but the postings change week to week.
- Other approved sites for postings include Dow (<https://corporate.dow.com/en-us/careers/jobs.html>), DuPont (<https://careers.dupont.com/us/en>) or Ashland (<https://www.ashland.com/>).

Enclosure 5: Installing LyX on Your Laptops

Before you install LyX, you must first install a TeX distribution. The program MikTeX is a common choice for this. Download and install MikTeX from <https://miktex.org/download>. Watch out for advertisers pushing other apps on this page. Make sure you download the file "basic-miktex-25.12-x64.exe." After downloading, run the executable as an administrator and accept all defaults during the installation. Near the end of the installation, MikTeX will ask you if you want to check for updates. This is a good idea but is not necessary now. LaTeX and LyX are constantly being updated and improved, but the updates take a while and is not necessary to get LyX up and running. You can always update MikTeX later.

After MikTeX has been installed, download the LyX installer from <https://www.lyx.org/Download>. Look for the file "LyX-251-Installer-1-x64.exe." After downloading, run the executable as an administrator and accept all defaults during the installation. The LyX installer will ask a series of questions about any packages it needs but cannot find. Make sure you install all of them. There can be many of these files and granting permission for each one can take a long time, so look for the checkbox that says, "always show this dialog" and deselect it (remove the check mark). The installation takes a while but should go smoothly as long as you accept all defaults and give permission to install any needed packages.